

Hypoxic Training and Empirical Assessment of Performance Output in Sports

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ABSTRACT

Notion of hypoxic training (HT) was evolved from Mexico Olympic Games held in 1968. HT protocols are: 1) Live high and train high (LHTH). 2) Live high and train low (LHTL). 3) Live low and train high (LLTH); live at sea level and to partake HT with artificial or natural altitude. Intermittent hypoxic training (IHT). And, 4) Intermittent hypoxic exposure (IHE). HT protocols with moderate altitude (~2800m) proven better results in most of the cases. Active training (LLTH) protocol(s) seems to be convenient protocol, consumes less time and minimal interface in routine training cycle of the player. Level of the player, phase of the training cycle, gender and iron status of the player needs to be taken care before embarking on the HT. Psychological, and field based game specific technical, tactical assessments may be the better instrumental of the performance assessment in addition to physiological and biochemical parameters after HT. HT must impart at pre-competition or competition phase as the beneficial effects last for few weeks. Systematic assessment of subjective and objective tests before and after HT may reveal the tangible benefits of HT.

Key words

Hypoxic training (HT), performance, physiologic implications and subjective and objective assessments

1. Rationale:

Oxygen is utmost vital element in the atmosphere, which maintains the life on the Earth (Chewy Gases 2015). Dry air at sea level is consisting of O₂, CO₂, N₂ and other inert gases, 20.98%, 0.04%, 78.06% and 0.92% respectively, and the total barometric pressure at sea level is 760 mm-Hg (Barrett et al., 2010). As we ascend higher, the air becomes lighter, atmospheric pressure tends to fall down and fractions of its composition relatively diminished.

Categorization of different altitudes as follows: >500 – 2000 m is considered as relatively low high altitude, > 2000 – 3000m is treated as moderate, >3000 – 5500m is considered as high and > 5500m is extremely high altitude (Saunders, Pyne, and Gore, 2009). Concept of high altitude training (HAT) was evolved as some of the athletes gained advantage of altitude in their performance in Mexico Olympic Games held in the year 1968, at 2300m altitude. Ever since, the coaches and the

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scientists have been striving to draw the potential benefit of high altitude training (Saunders et al., 2013). In last few years, different altitude training methods have been adopted to enhance performance of various endurance as these methods are legal, safe and easy to implement. Artificial altitude training methods had initially developed and applied to enhance the performance of athletes in Scandinavia and USA in early 1990s (Gregoire P. Millet et al., 2010; Park et al., 2016). With few choices of legal aspects, the World Anti-Doping Agency (WADA), allows the athletes to undergo hypoxic training (HT) under biological passport criterion (The World Anti-doping agency 2018; Zorzoli 2011).

2. Hypoxic Training (HT) Protocols:

Hypoxic training (HT) has different modalities of training regimes and these are 1) Live high and train high (LHTH); stay in high altitude/hypoxic chamber and training in high altitude with pre-decided load. It is an active protocol; 2) Live high and train low (LHTL); stay in high altitude/hypoxic chamber and train at sea level, and is a passive protocol and 3) Live low and train high (LLTH); live at sea altitude or near the sea level and to partake HT with artificial or natural altitude. Intermittent hypoxic training (IHT). And Intermittent hypoxic exposure (IHE); an altitude training exposure to extreme alternative hypoxic and normoxic states by using a hypoxic device (individual mask). Further, the principle of the HT may be a natural (natural hypobaric), or

artificial chamber (artificial hypobaric), wherein a chamber creates the simulated high altitude with reduced total barometric pressure. Lastly, normobaric hypoxic chamber (artificial normobaric), it essentially, create normobaric hypoxic environment in artificial chamber with manipulation of only partial pressure of oxygen (PO_2) inside the chamber (Grégoire P Millet et al. 2013; Park et al. 2016; Sinex and Chapman 2015; Wilber 2007, 2011)

3. Better HT Protocol:

Of many methods of HT protocols, LHTL training method with altitude of 2500-3000m ($FiO_2=14.6\%$) is the widely accepted protocol across the globe. Moderate altitude of 2800m of passive hypoxic exposure (LHTL), yields maximum physiological, and performance benefits to the athletes. Erythropoietin (EPO) response escalates ~ 92% after 24h of post exposures and other essential physiological markers are substantially high at this range of HT. It has been well documented that various duration of hypoxic exposures in different publications. However, 8-12h/day exposure for 21-28 days protocol with moderate altitude (~ 2800m) is highly recommended. Nevertheless, endurance players are maximally benefitted following HT (Chapman 2013; Girard et al. 2013; Humberstone-Gough et al. 2013; Saugy et al. 2014; Sinex Chapman 2015 and Adhikari S. et al. 2017). Currently, intermittent hypoxic training (IHT) has also been emerged out as beneficial to the athletes in tropical countries (Miłosz

Czuba et al. 2013; Truijens et al. 2003). However, even repeated sprint trainings in hypoxia has been suggested to be a better training regimen than IHT (Faiss, Girard, and Millet 2013). A number of evidences suggest that both active and passive protocols have beneficial effects on various physiologic parameters (Adhikari S. et al. 2017, Kumar GK, Klein JB. 2004.). Nevertheless, as IHT requires minimum time exposure to artificial high altitude training (AHT) and the actual routine field training will not disrupt in tentire HT (Hamlin et al. 2010). Eventually, time of imparting HT during the training cycle also be a critical as HT effects may last up to 25 – 35 days (Sinex and Chapman 2015). Hence, HT to be planned at pre-competition or competition period of the training cycle.

4. Iron Supplementation During HM:

It has well been established that, improvement of the haematological markers are the key beneficial parameters to HT. Physiologic benefits, enhancement of performance of HT may be amplified with iron supplementation (Stray-Gundersen, Chapman, and Levine 2001). Very often, female athletes probably have inadequate iron stores; non anaemic iron deficiency (NAID; serum ferritin < 20 µg/mL) (Stray-Gundersen, Chapman, and Levine 2001; Torti and Torti 2002). Ferritin is a storage protein; iron is stored in the ferric form associated with hydroxide and phosphate anions (Torti and Torti 2002).

A study reported that ferritin levels have significantly dropped followed by a 3 week male swimming training (Roberts and Smith 1992). Pre iron supplementation protocol were adopted in most of the HT studies (Courtney 2001; Govus et al. 2015).

5. Physiologic Implications to HT:

Ultimate goal of HT (different methods and protocols) is to improve the physiological markers and performance (Humberstone-Gough et al. 2013; Saugy et al. 2014, 2016). Many research outcomes have proven that HT provokes multilevel physiological responses. These responses begins from cellular level and then to performance. HT initially elicits hypoxic induced factor (HIF) and then cascades multilevel bio-signalling process at cellular level. Increased EPO secretion have been reported in response to HT by many findings (Milosz Czuba et al. 2011; Ge et al. 2002; Gregoire P. Millet et al. 2010; Muraoka and Gando 2012). Augmentation of aerobic capacity; volume of maximum oxygen consumption during exercise (VO₂max) has been observed in maximum studies (Maciejczyk, Sudoł, and Szyguła 2012; Park et al. 2016; Saugy et al. 2016). Improvement of haematological markers are the most common physiological advantage to the HT (Milosz Czuba et al. 2013; Milosz Czuba et al. 2011; Gregoire P. Millet et al. 2010; Vargas Pinilla 2014). Lactate threshold, running economy are also part of performance enhancer apart from the

haematological benefit. Number of studies have shown that lactate threshold and running economy are significantly enhanced after HT (Levine and Stray-Gundersen 1997; Saunders et al. 2013). HT enhances the muscle buffering capacity by clearing lactate during high intensity exercise (Muraoka and Gando 2012). Studies also reported on significant enhancement of anaerobic capacity of the athletes followed by the HT but nonetheless, not limited to LHTL (Hamlin et al. 2010; Levine and Stray-Gundersen 1997; Gregoire P. Millet et al. 2010).

6. Assessment of the Athlete Performance:

Available database of research implies enhancement of performance has been proven followed by various HT protocols. Having mentioned that, the performance improvement demonstrated in terms of haematological, lactate threshold, running economy, aerobic capacity, anaerobic capacity and other physiological benefits. There is no empirical research proven to be improvement of categorical performance followed by HT. Since, performance is a wide entity which include array of domains. Nonetheless, improvement of physiological parameters may not merely reflect the performance at real situation of the competition. However, gaining physiological benefit is one of many factors to be involved in performance. Further, improvement of performance of highly trained elite level players through HT is really challenging as they already

reached to their peak performance.

Multidimensional approach of assessment of various components may reveal cumulative output of the performance followed by HT. Psychological assessment like athlete subjective self-reported measures may also be useful, apart from physiologic and biochemical parameters followed by a HT (Saw, Main, and Gastin 2016). The compound assessment of psychological parameters may reveal how the player responds by means of well being, mood state and recovery (training response), in overall response to HT. Field based, game specific, technical, tactical and physical assessments would essentially provide the actual real-time benefits followed by HT. Suggestions from expert/scientist of training methods may simplify the type of training to be imparted in HT, and it depends on type of sport, level of the player and phase of the training cycle of group/individual.

7. Conclusion:

Hypoxia has great impact on systemic physiology (Kumar et. al., 2014). Molecular oxygen participate / regulates the various bio-signalling processes at cellular level (Kumar G.K. & Klein J.B., 2004). In spite of the availability of various training modalities, there is a widespread interest among researchers to find out which method is more suitable for a specific sports population. However, choosing the right HT is depends on multiple factors. Structured, passive or active HT may facilitate enhancement of

the performance of the player. Level of the player, phase of the training cycle, gender and iron status of the player needs to be taken care before embarking on the HT. Psychological, and field based game specific technical, tactical assessments may be better instrumental in performance assessment, in addition, to the physiological and biochemical

parameters followed by HT. Systematic assessment of subjective and objective tests before and after HT may reveal the actual benefits of HT. Moreover, HT can be used as means and methods of training for the athletes and it can be incorporated in the periodization of the training plan.

Bibliography:

- Adhikari Snehusu, Sreeman Radhica, Saraswat Sanjay, Medabala Tambi, Singh Wazir. 2017.** "Does 'live high-train low' hypoxic training really improves endurance capacity of long distance runners?" *European Respiratory Journal* 50 (SUPPL. 61.): PA2269; DOI: 10.1183/1393003.congress-2017.PA2269.
- Barrett, Kim, Heddwen Brooks, Scott Boitano, and Susan Barman. 2010.** Ganong's review of medical physiology Ganong's Review of Medical Physiology.
- Chapman, Robert F. 2013.** "The Individual Response to Training and Competition at Altitude." *British Journal of Sports Medicine* 47(SUPPL. 1): 1–7.
- Courtney, Rosalba. 2001.** "Intermittent Hypoxic Training." *International Wellbeing* (83).
- Czuba, Milosz et al. 2013.** "The Effects of High Intensity Interval Training in Normobaric Hypoxia on Aerobic Capacity in Basketball Players." *Journal of Human Kinetics* 39(1): 103–14.
- Czuba, Milosz et al. 2011.** "The Effects of Intermittent Hypoxic Training on Aerobic Capacity and Endurance Performance in Cyclists." *Journal of Sports Science and Medicine* 10(1): 175–83.
- Faiss, Raphaël, Olivier Girard, and Grégoire P. Millet. 2013.** "Advancing Hypoxic Training in Team Sports: From Intermittent Hypoxic Training to Repeated Sprint Training in Hypoxia." *British Journal of Sports Medicine* 47(SUPPL. 1): 2–9.
- Ge, Ri-Li et al. 2002.** "Determinants of Erythropoietin Release in Response to Short-Term Hypobaric Hypoxia." *Journal of Applied Physiology* 92(6): 2361–67.
- Girard, Olivier et al. 2013.** "Position Statement-Altitude Training for Improving Team-Sport Players' Performance: Current Knowledge and Unresolved Issues." *British Journal of Sports Medicine* 47(SUPPL. 1).
- Govus, Andrew D. et al. 2015.** "Pre-Altitude Serum Ferritin Levels and Daily Oral Iron Supplement Dose Mediate Iron Parameter and Hemoglobin Mass Responses to Altitude Exposure." *PLoS ONE* 10(8): 1–11.
- Hamlin, M. J. et al. 2010.** "Effect of Intermittent Hypoxic Training on 20 Km Time Trial and 30 S Anaerobic Performance." *Scandinavian Journal of Medicine and Science in Sports* 20(4): 651–61.

- Humberstone-Gough, Clare E. et al. 2013.** "Comparison of Live High: Train Low Altitude and Intermittent Hypoxic Exposure." *Journal of Sports Science and Medicine* 12(3): 394–401.
- Kumar GK, Klein JB. 2004.** "Analysis of expression and post-translational modification of proteins during hypoxia." *J Appl Physiol*. 96:1178–86.
- Kumar Santosh, Sharma Priyanka, Bansal Anju, Sharma Prakash C., and Aggarwal Kamal K. 2014.** "Hypobaric hypoxia-mediated protein expression in plasma of susceptible & tolerant rats" *Indian J Med Research* 140(6):756–765.
- Levine, B D, and J Stray-Gundersen. 1997.** "'Living High-Training Low': Effect of Moderate-Altitude Acclimatization with Low-Altitude Training on Performance." *Journal of applied physiology* (Bethesda, Md. : 1985) 83(1): 102–12.
- Maciejczyk, Marcin, Grzegorz Sudół, and Zbigniew Szygula. 2012.** "Influence of Hypoxia Training on the Aerobic Capacity of an Elite Race Walker." *Human Movement* 13(4): 360–66.
- Millet, Gregoire P. et al. 2010.** "Combining Hypoxic Methods for Peak Performance." *Sports Medicine* 40(1): 1–25.
- Millet, Grégoire P, Raphaël Faiss, Franck Brocherie, and Olivier Girard. 2013.** "Hypoxic Training and Team Sports : A Challenge to Traditional Methods ?" 47(December): 6–7.
- Muraoka, Isao, and Yuko Gando. 2012.** "Effects of the 'Live High-Train High' and 'Live High-Train Low' Protocols on Physiological Adaptations and Athletic Performance." *J Phys Fitness Sports Med* 1(3): 447–55.
- Park, Hun-young et al. 2016.** "The Effects of Altitude/hypoxic Training on Oxygen Delivery Capacity of the Blood and Aerobic Exercise Capacity in Elite Athletes – a Metaanalysis." *Journal of Exercise Nutrition & Biochemistry* 20(1): 15–22.
- Roberts, D, and D J Smith. 1992.** "Training at Moderate Altitude - Iron Status of Elite Male Swimmers." *Journal of Laboratory and Clinical Medicine* 120(3): 387–91.
- Saugy, Jonas J et al. 2014.** "Comparison of 'Live High-Train Low' in Normobaric versus Hypobaric Hypoxia."
- Saugy, Jonas J. et al. 2016.** "Same Performance Changes after Live High-Train Low in Normobaric vs. Hypobaric Hypoxia." *Frontiers in Physiology* 7(APR): 1–10.
- Saunders, Philo U., Laura A. Garvican-Lewis, Walter F. Schmidt, and Christopher J. Gore. 2013.** "Relationship between Changes in Haemoglobin Mass and Maximal Oxygen Uptake after Hypoxic Exposure." *British Journal of Sports Medicine* 47(SUPPL. 1).
- Saunders, Philo U., David B. Pyne, and Christopher J. Gore. 2009.** "Endurance Training at Altitude." *High Altitude Medicine & Biology* 10(2): 135–48.
- Saw, Anna E., Luana C. Main, and Paul B. Gastin. 2016.** "Monitoring the Athlete Training Response: Subjective Self-Reported Measures Trump Commonly Used Objective Measures: A Systematic Review." *British Journal of Sports Medicine* 50(5): 281–91.
- Sinex, Jacob A., and Robert F. Chapman. 2015.** "Hypoxic Training Methods for Improving Endurance Exercise Performance." *Journal of Sport and Health Science* 4(4): 325–32.
- Stray-Gundersen, J, R F Chapman, and B D Levine. 2001.** "'Living High-Training Low' altitude

- Training Improves Sea Level Performance in Male and Female Elite Runners." *Journal of applied physiology* (Bethesda, Md. : 1985) 91: 1113–20. "Sustaining Life on Planet Earth: Metalloenzymes Mastering Dioxygen and Other Chewy Gases." 2015. 15: 25707464. <http://link.springer.com/10.1007/978-3-319-12415-5>.
- The World Anti-doping agency. 2018.** "Standard Prohibited List 2018." (JANUARY).
- Torti, Frank M., and Suzy V. Torti. 2002.** "Regulation of Ferritin Genes and Protein." *Blood* 99(10): 3505–16.
- Truijens, M. J., H. M. Toussaint, J. Dow, and B. D. Levine. 2003.** "Effect of High-Intensity Hypoxic Training on Sea-Level Swimming Performances." *Journal of Applied Physiology* 94(2): 733–43.
- Vargas Pinilla, Olga Cecilia. 2014.** "Exercise and Training at Altitudes: Physiological Effects and Protocols." *Ciencias de la Salud* 12(1): 115–30.
- Wilber, Randall L. 2007.** "Introduction to Altitude/hypoxic Training Symposium." *Medicine and Science in Sports and Exercise* 39(9): 1587–89. 2011. "Application of Altitude/hypoxic Training by Elite Athletes." *Journal of Human Sport and Exercise* 6(2): 271–86.
- Zorzoli, Mario. 2011.** "Biological Passport Parameters." *Journal of Human Sport and Exercise* 6(2): 205–17.